2 **(a)** State Newton's first law of motion.

.....

.....

..... [1]

(b) Block X of mass 2.0 kg and block Y of mass 4.0 kg, initially at rest, are connected by a light inextensible string that passes over two frictionless light pulleys P_1 and P_2, with one end of the string fixed to the ceiling as shown in Fig. 2.1.

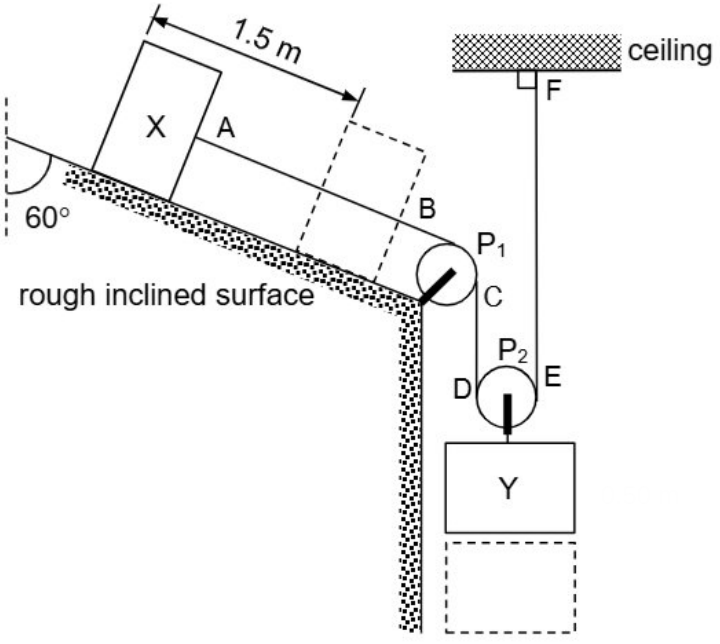


Fig. 2.1

AB is parallel to the inclined surface and CD is parallel to EF.

Both blocks subsequently start to move. The inclined surface exerts a constant frictional force of 5.0 N on block X such that the acceleration of block X is twice the acceleration of block Y.

(i) Calculate the work done by the frictional force when block X moves 1.5 m along the inclined surface.

work done = ... J [1]

(ii) Hence or otherwise, determine the total kinetic energy of the system when block X moves 1.5 m along the inclined surface.

kinetic energy = ... J [2]

(iii) Fig. 2.2 shows a body of system consisting of the string, pulley P_2 and block Y.

On Fig. 2.2, draw and label the forces acting on the body.

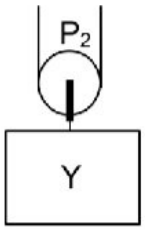


Fig. 2.2

[1]

(iv) Calculate the tension T in the string.

T = ... N [3]